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In [1]: # Polynomial fit
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
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In [2]: df = pd.read_csv("Advertising.csv").iloc[:,1:]
df.head()
df_small = df.iloc[0:100,[0,-1]]
df_small
```

Out[2]:

	TV	Sales
0	230.1	22.1
1	44.5	10.4
2	17.2	9.3
3	151.5	18.5
4	180.8	12.9
...	...	...
95	163.3	16.9
96	197.6	11.7
97	184.9	15.5
98	289.7	25.4
99	135.2	17.2

100 rows × 2 columns

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In [3]: xInput = df_small.iloc[np.r_[10:100],0].values
yInput = df_small.iloc[np.r_[10:100],-1].values
xTest = df_small.iloc[0:10,0].values
yTest = df_small.iloc[0:10,1].values
```

```
In [6]: # Polynomial fit
class polyFit:
    def __init__(self):
        self.n = 0
        self.x = 0
        self.y = 0

    def modelFit(self, xInput, n):
        for i in range(1, n+1):
            xSub = xInput.reshape(-1,1) ** i
            if (i == 1):
                x = np.hstack( [ xSub, np.ones((len(xInput),1)) ] ) # x matrix for t
            else:
                x = np.hstack([xSub, x]) # x matrix for training data
```

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    return x

def minFind(self, x, y):
    # Apply normal equation to find minimum for slopes and y-intercept paramete
    xtx = x.T @ x
    w = np.linalg.inv(xtx) @ x.T @ y
    return w

def polyPlot(self, w, xInput, yInput, xTest, yTest):
    wReversed = w[::-1]
    xPlot = np.arange(0, max(xInput)+100, 0.1)
    for i in range(0, len(wReversed)):
        if (i == 0):
            yPlot = wReversed[i]
        if (i > 0):
            yPlot = wReversed[i] * xPlot ** i + yPlot

    dist = 0
    for j in range(0, xTest.size):
        idx = (np.abs(xPlot - xTest[j])).argmin()
        dist = (yTest[j] - yPlot[idx]) + dist
    avg = dist / xTest.size
    print("Average residual distance for test data: ", avg)

    # plot
    plt.scatter(xInput, yInput, marker = "*", color = 'r') # training data
    plt.scatter(xTest, yTest, marker = 's', color = 'g') # test data
    plt.title(print("Polynomial fit: degree = ", len(w)-1))
    plt.plot(xPlot, yPlot, color = 'b', linestyle = '--')
    plt.show()

```

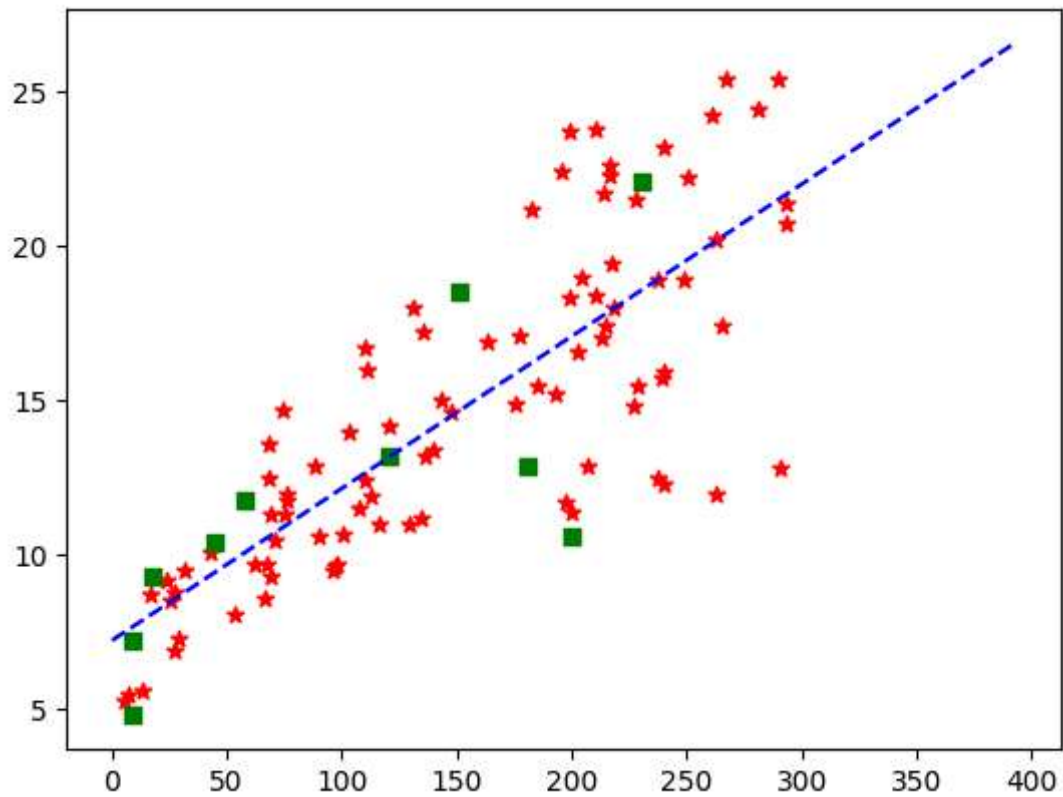
```

In [7]: for i in range(1, 7):
        p = polyFit()
        x = p.modelFit(xInput, i) # returns x matrix fit to polynomial
        w = p.minFind(x, yInput) # returns minimum values for features in polynomial
        p.polyPlot(w, xInput, yInput, xTest, yTest)

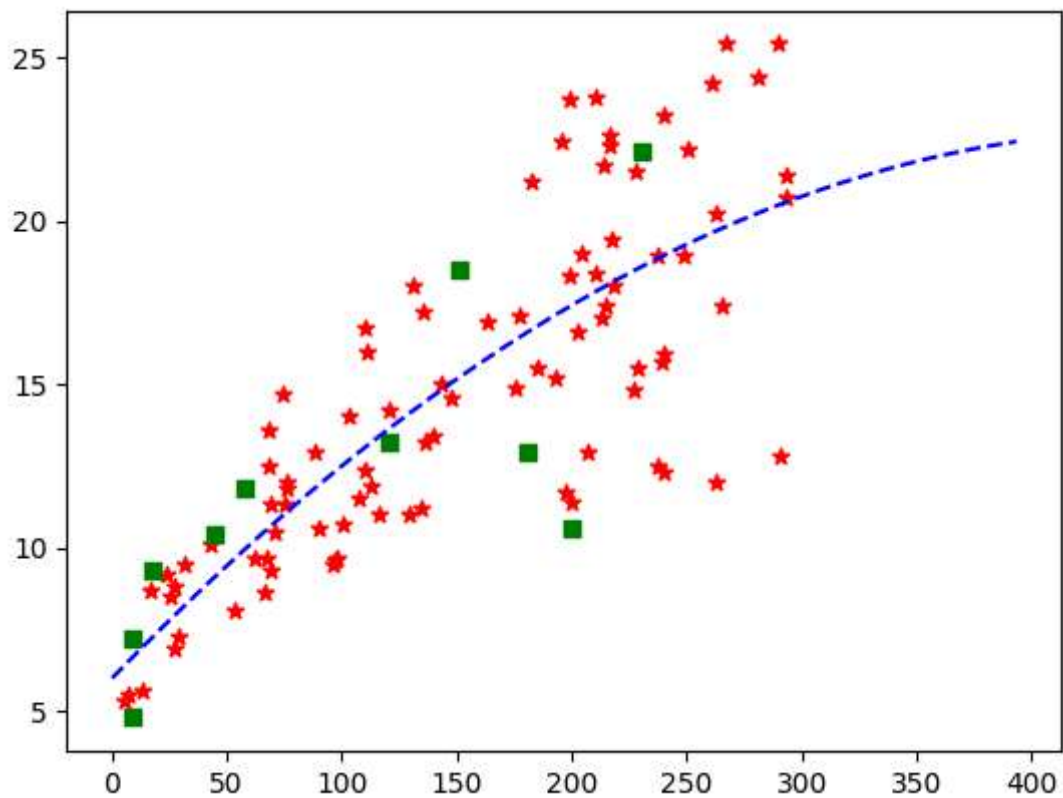
```

Average residual distance for test data: -0.17269440008445072

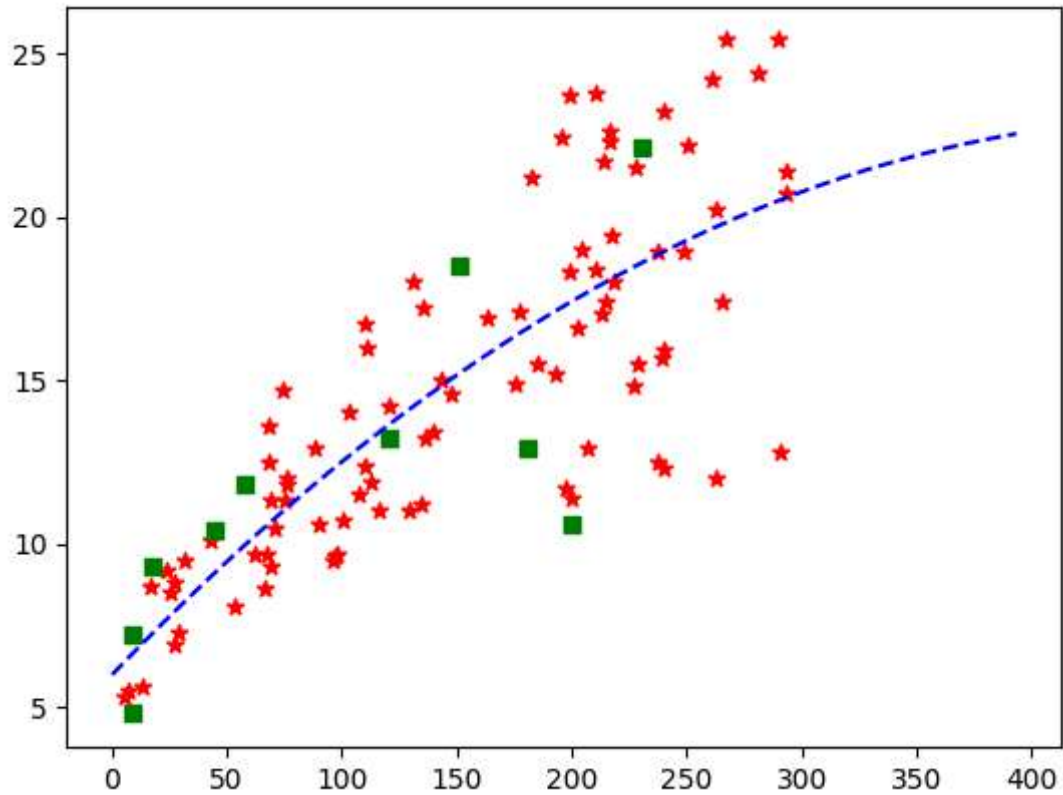
Polynomial fit: degree = 1



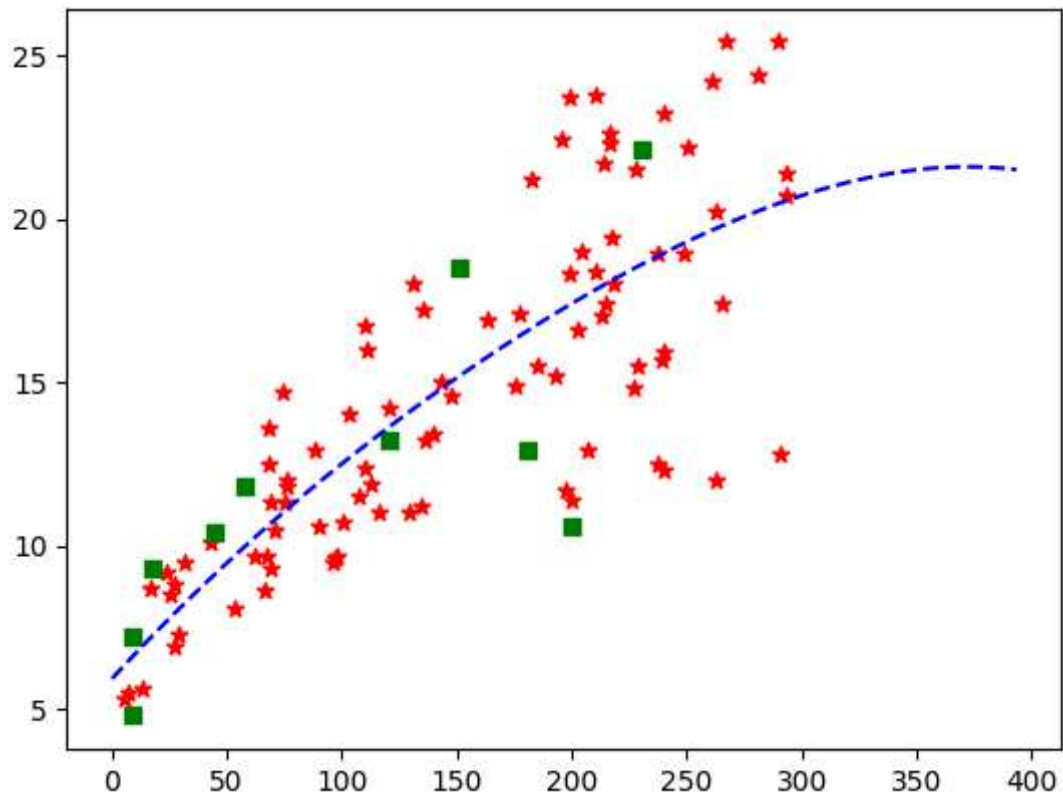
Average residual distance for test data: -0.01705062133364388
Polynomial fit: degree = 2



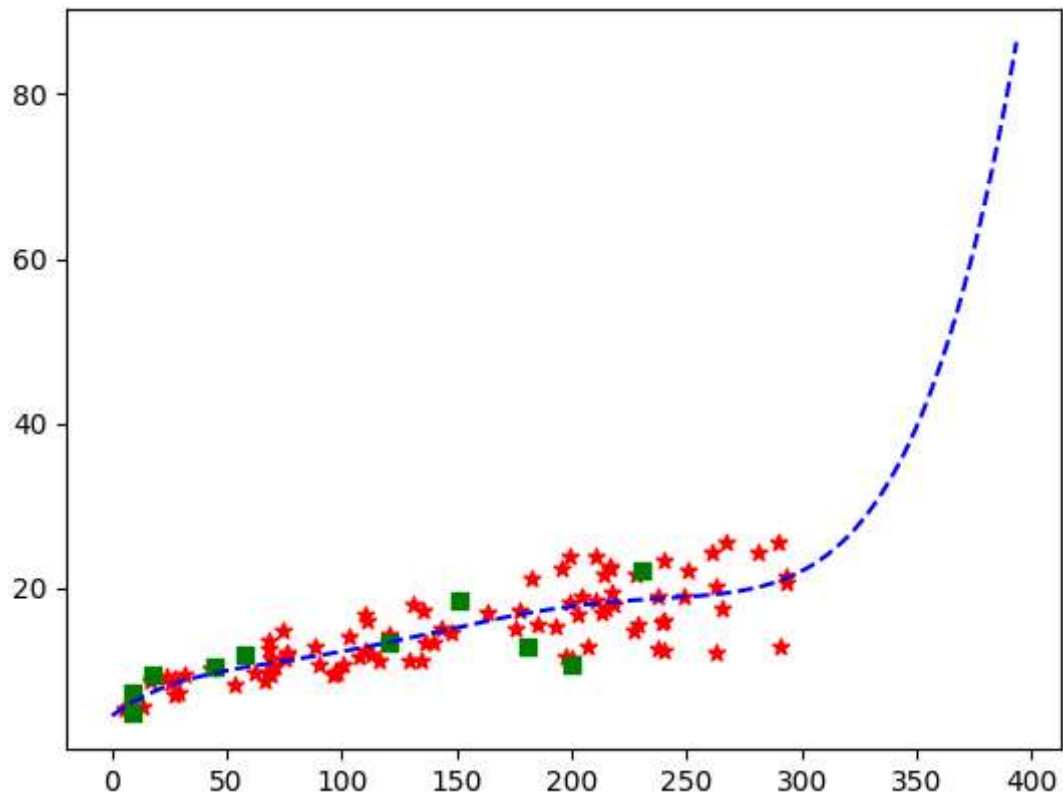
Average residual distance for test data: -0.013693598377994488
Polynomial fit: degree = 3



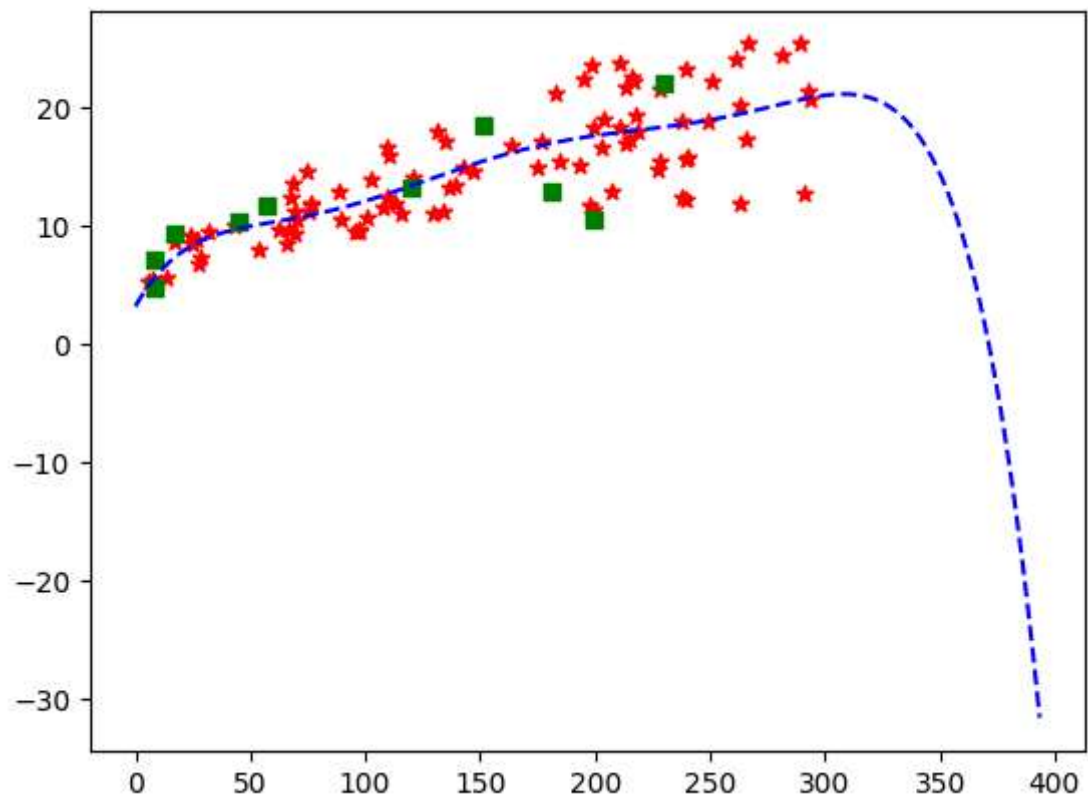
Average residual distance for test data: -0.00637595920914027
Polynomial fit: degree = 4



Average residual distance for test data: -0.0015716197135298593
Polynomial fit: degree = 5



Average residual distance for test data: -0.0045155048584574866
Polynomial fit: degree = 6



In []: